"Hello, IPC"

Connecting Rust and Zephyr

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Nairobi Linux User Group
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SPEAKER

Hello, Chrispine

- Embedded Systems Engineer
- Zephyr RTOS, Rust, Python, and EMS
- Open-source contributor
- Today: a live embedded demo, plus two real pull requests

AT A GLANCE

✓	zephyrproject-rtos/zephyr watchdog driver Cortex-M fix	#118106
●	pymodbus-dev/pymodbus socket reconnect fix	#3000
✓	zephyrproject-rtos/zephyr BL350 board support	#113302
✓	zephyrproject-rtos/zephyr-lang-rust Rust blinky sample fix	#132
●	PyPSA/PyPSA pandas dtype fix	#1151

AGENDA

We'll cover

01 Why chips have more than one brain

02 Zephyr and Rust, in one sentence each

03 The idea: a coprocessor is just a file

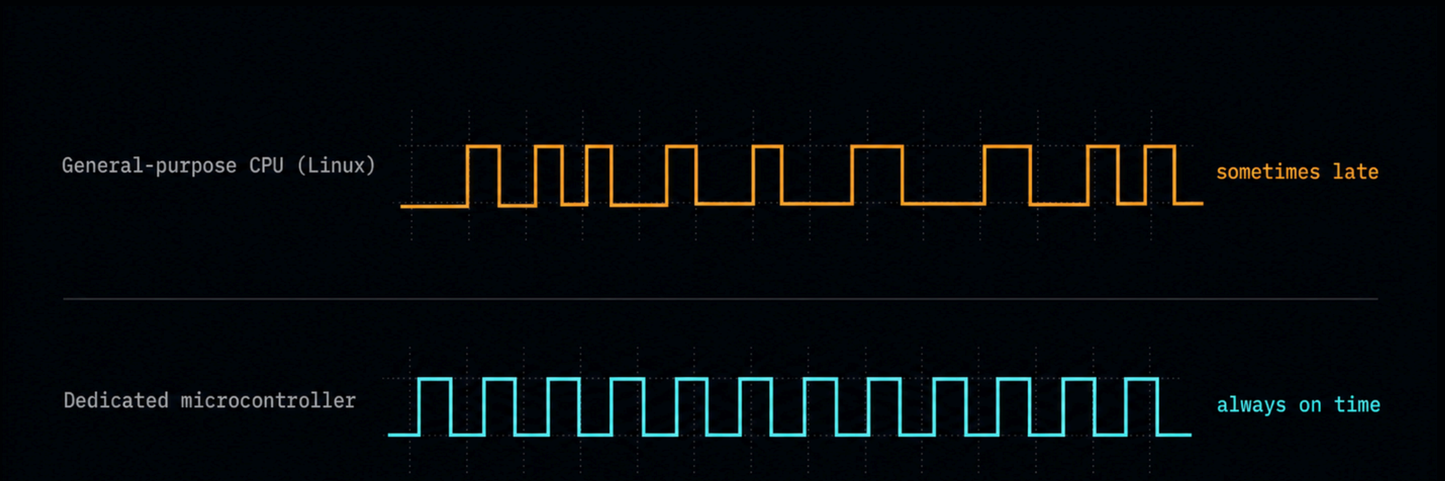
04 Live demo on real hardware

05 What broke, and what it taught me

06 Two pull requests that got this here

Linux is not good at promises about time

- Linux schedules your process when it gets around to it
- “Usually within a millisecond" ≠ "always"
- Some jobs need guarantees: hold a pulse for 2 ms, sample every 500 µs



TAKEAWAY

Pair a general-purpose core with a small microcontroller that runs nothing else.

Two runtimes, one chip

ZEPHYR

- Small open-source RTOS
- Runs on the microcontroller core
- Predictable, real-time-oriented
- Core written in C, with growing Rust support

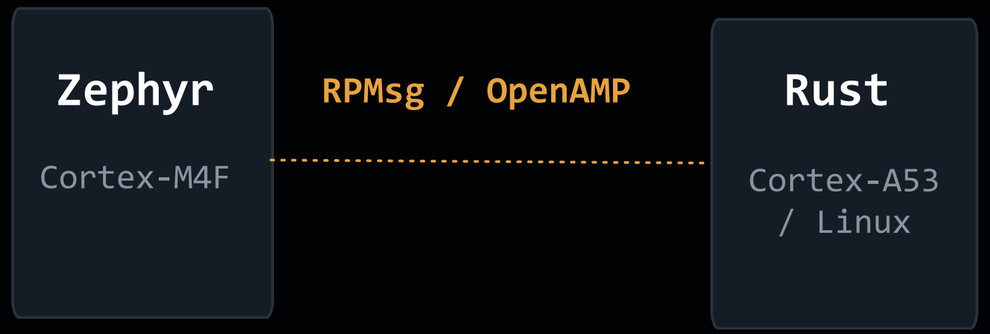
Zephyr®

RUST (on Linux)

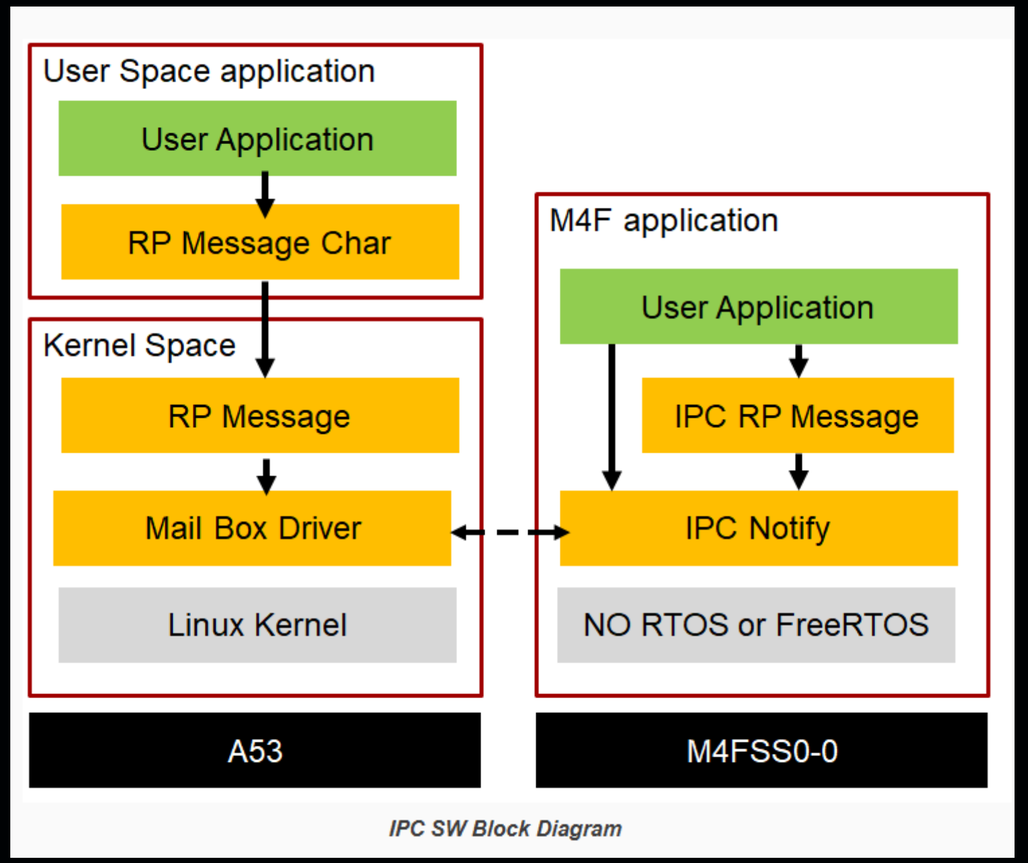
- Memory-safe systems language
- Runs on the Linux application core
- No garbage collector, fast, reliable
- Talks to the world like a normal app

What is IPC, really?

Inter-Process Communication: two programs, exchanging messages



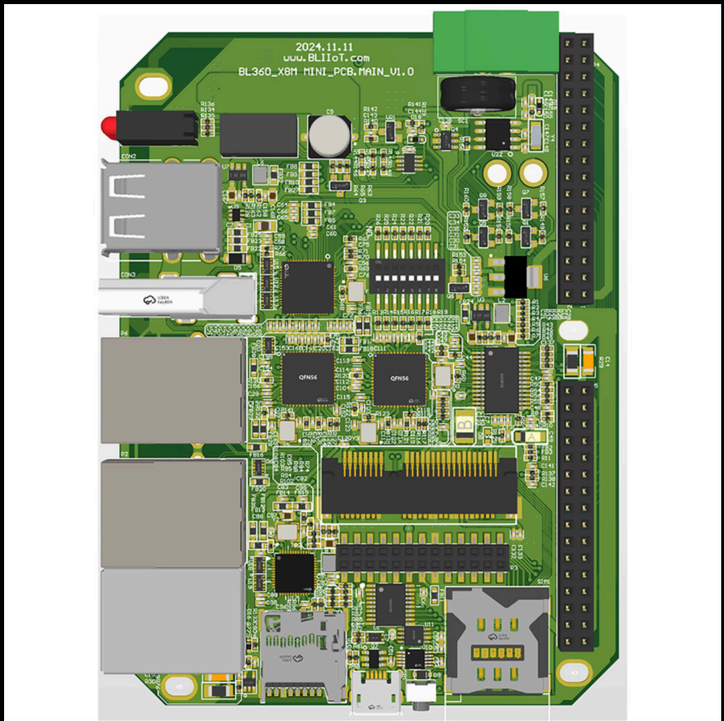
Same chip. Shared memory. Two completely different worlds, one conversation.



Source: software-dl.ti.com

The board

- Industrial embedded controller



- I upstreamed this board into mainline Zephyr

BOARD SPEC

am62x_m4_bl350

SOC	TI AM6254
LINUX CORES	4x Cortex-A53 @ 1.4 GHz
RTOS CORE	1x Cortex-M4F @ 400 MHz
SHARED IPC REGION	1 MB (OpenAMP / RPMsg vrings)
STATUS	Maintained, mainline Zephyr

Source: Zephyr documentation

On Linux, the coprocessor is a file

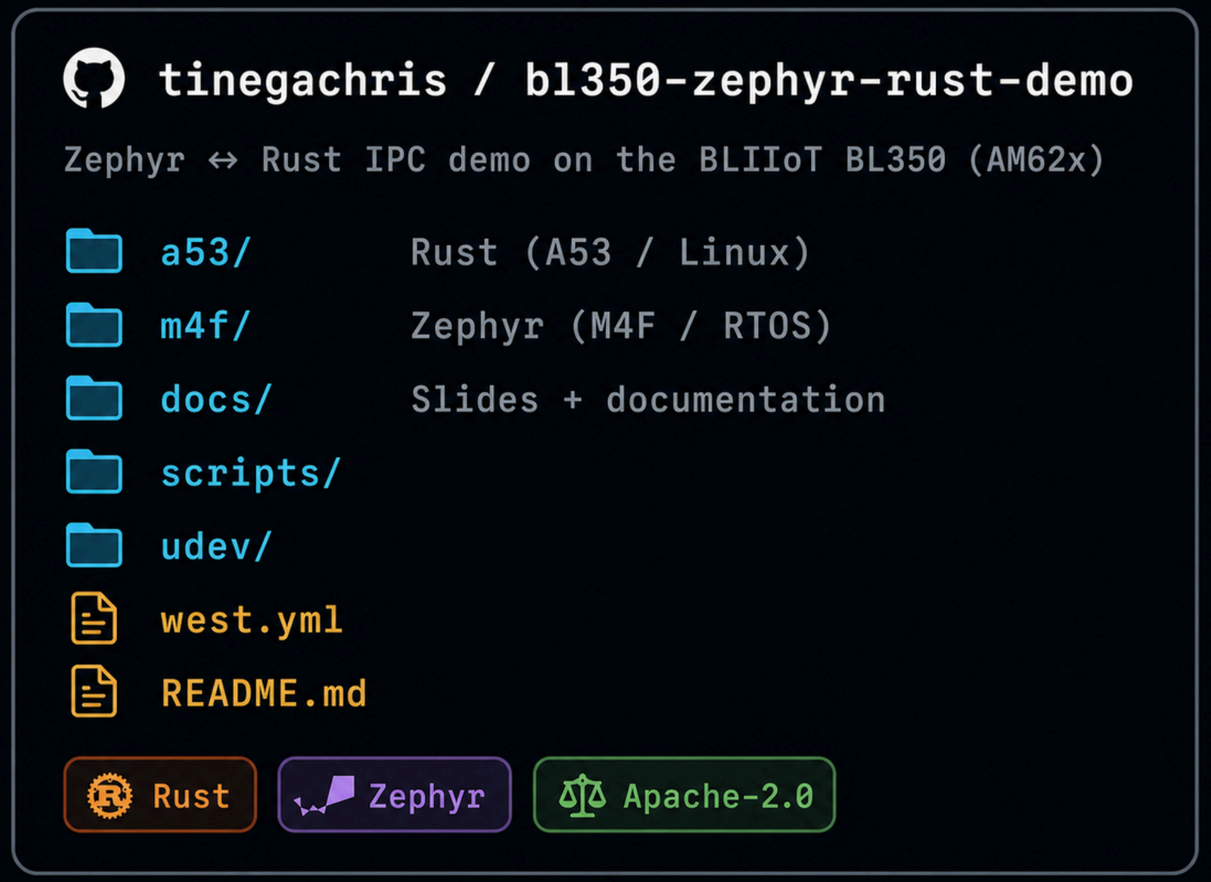
```
let mut endpoint = OpenOptions::new()
    .read(true).write(true)
    .open(path)?;

endpoint.write_all(b"Hello from Rust")?;
let n = endpoint.read(&mut buf)?;
```

No driver. No ioctl. No crate.

std + a character device. That's the whole program.

Let's watch it talk

- Rust on Linux (A53) sends a greeting over RPMsg
- Zephyr on the coprocessor (M4F) replies with its uptime
- Verified live on the real am62x_m4_bl350 board

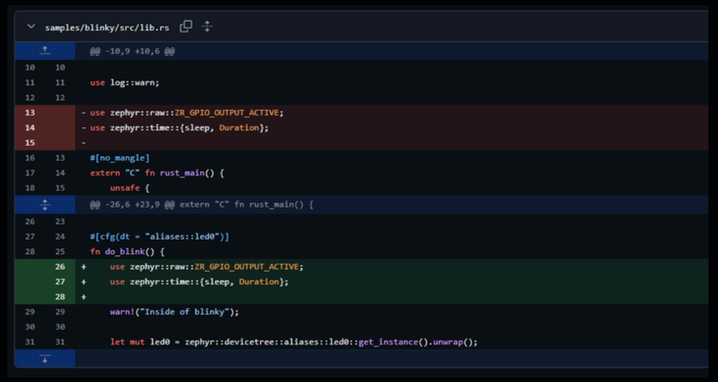


It started with moving a few lines

- Upstream PR in the Rust + Zephyr ecosystem
- samples: rust: blinky: Use conditional imports to fix compiler warnings
- Literally just moved imports and fixed a warning
- Proof that your first contribution doesn’t have to be huge

RUSTLANG FIRST PR

zephyrproject-rtos/zephyr#132
Use conditional imports to fix compiler warnings



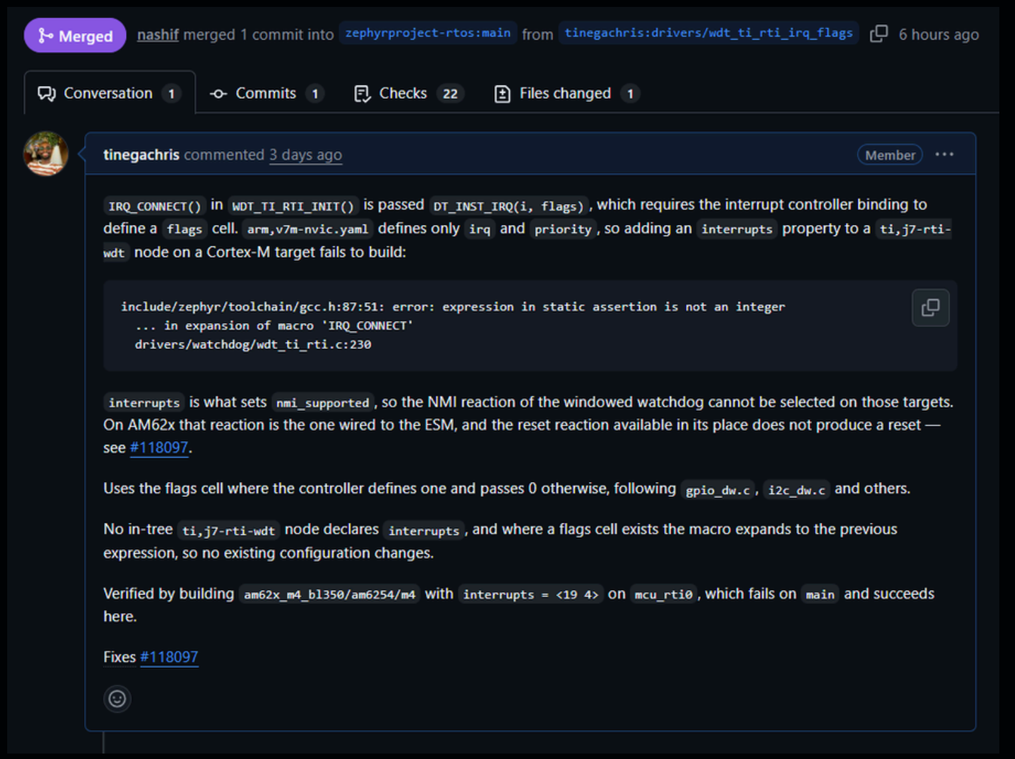
Tiny cleanups still matter,
and they teach you the review process.

Now it fixes real hardware

- Recent merged PR in Zephyr upstream
- drivers: watchdog: wdt_ti_rti: fix build with interrupts on Cortex-M
- Makes the AM62x windowed watchdog build correctly with interrupts
- Same contributor now fixing real bugs

MOST RECENT MERGE

zephyrproject-rtos/zephyr#118106
Watchdog fix on Cortex-M



Merged nashif merged 1 commit into zephyrproject-rtos:main from tinegachris:drivers/wdt_ti_rti_irq_flags 6 hours ago

Conversation 1 Commits 1 Checks 22 Files changed 1

tinegachris commented 3 days ago Member

`IRQ_CONNECT()` in `WDT_TI_RTI_INIT()` is passed `DT_INST_IRQ(i, flags)`, which requires the interrupt controller binding to define a `flags` cell. `arm,v7m-nvic.yaml` defines only `irq` and `priority`, so adding an `interrupts` property to a `ti,j7-rti-wdt` node on a Cortex-M target fails to build:

```
include/zephyr/toolchain/gcc.h:87:51: error: expression in static assertion is not an integer
  ... in expansion of macro 'IRQ_CONNECT'
  drivers/watchdog/wdt_ti_rti.c:230
```

`interrupts` is what sets `nmi_supported`, so the NMI reaction of the windowed watchdog cannot be selected on those targets. On AM62x that reaction is the one wired to the ESM, and the reset reaction available in its place does not produce a reset — see #118097.

Uses the flags cell where the controller defines one and passes 0 otherwise, following `gpio_dw.c`, `i2c_dw.c` and others.

No in-tree `ti,j7-rti-wdt` node declares `interrupts`, and where a flags cell exists the macro expands to the previous expression, so no existing configuration changes.

Verified by building `am62x_m4_bl350/am6254/m4` with `interrupts = <19 4>` on `mcu_rti0`, which fails on `main` and succeeds here.

Fixes #118097

Start where I started

One typo fix, one missing doc, one small driver bug.

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